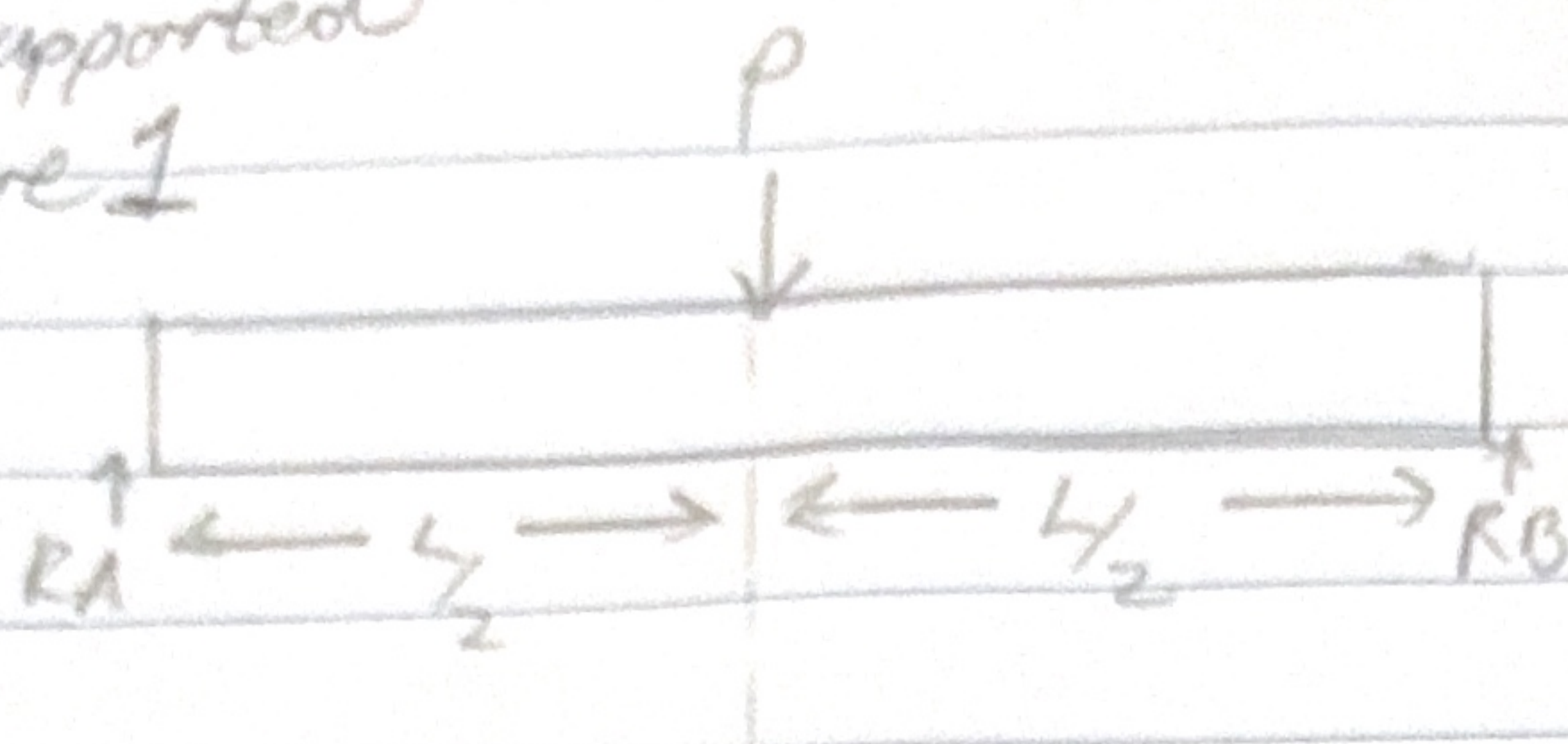


* Simply supported
Structure 1



$$R_A + R_B - P = 0$$

$$\sum F_y = 0$$

$$\sum F_x = 0$$

$$\sum M = 0$$

$$\sum F_y = R_A + R_B - P$$

$$R_A + R_B - P = 0$$

$$\sum M_A = R \left(P \times \frac{L}{2} \right) - (R_B \times L) = 0$$

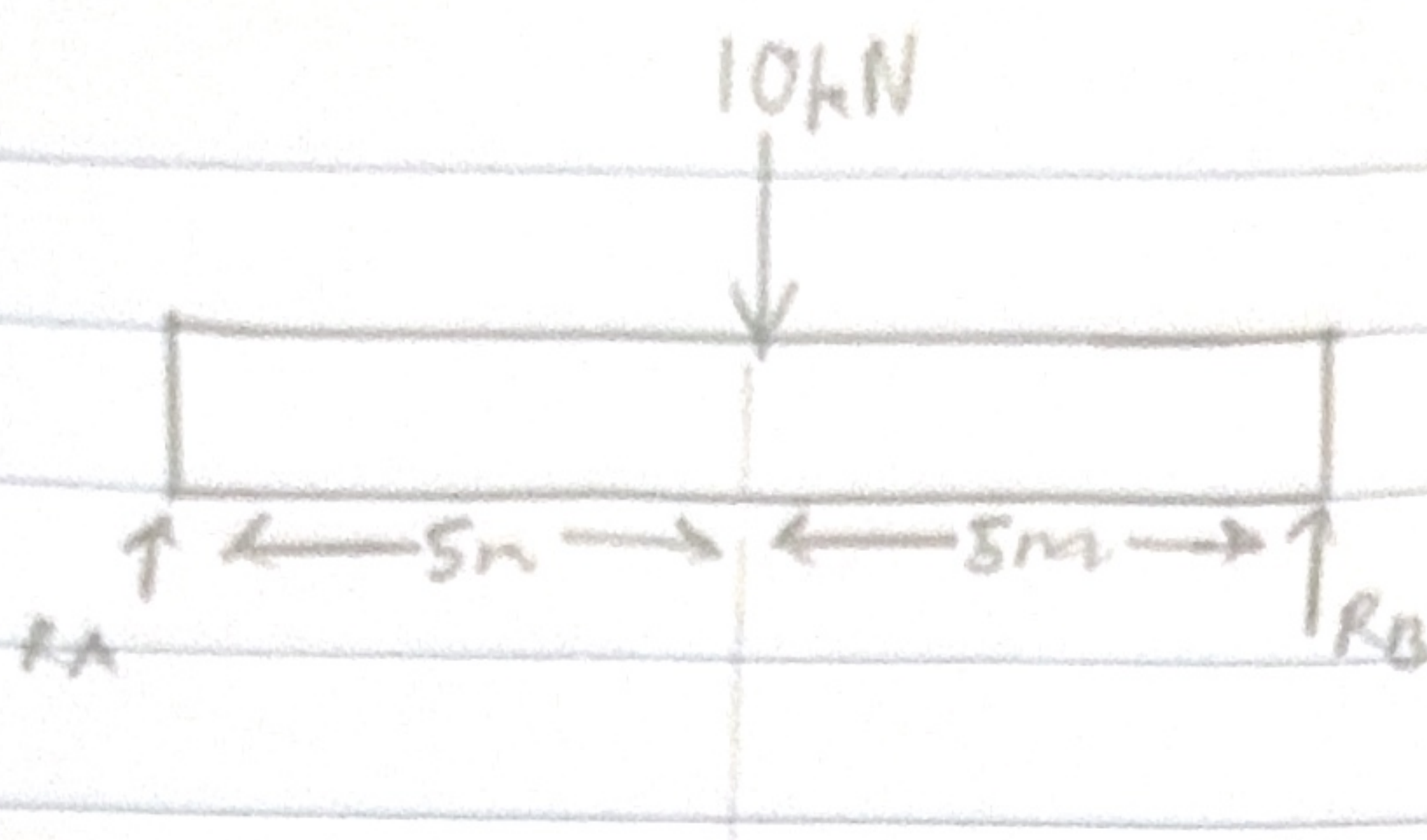
$$R_B L = \frac{P L}{2}$$

$$R_B = \frac{P}{2}, \text{ since } P \text{ is the load force}$$

$$R_B = \frac{F}{2}$$

$$R_A + \frac{P}{2} - P = 0 \Rightarrow R_A = \frac{F}{2}$$

$$R_A = R_B = \frac{F}{2} / \frac{P}{2}$$



$$R_A + R_B - 10 = 0$$

$$\sum F = 0, \sum F_x = 0, \sum F_y = 0$$

$$\sum M = 0, \sum M_A = 0, \sum M_B = 0$$

$$\begin{aligned} \sum M_A &= (10 \times 5) - (R_B \times 10\text{m}) = 0 \\ &= 50 - 10R_B = 0 \end{aligned}$$

$$R_B = \frac{50}{10} = 5\text{kN}$$

$$R_A + R_B - 10\text{kN} = 0$$

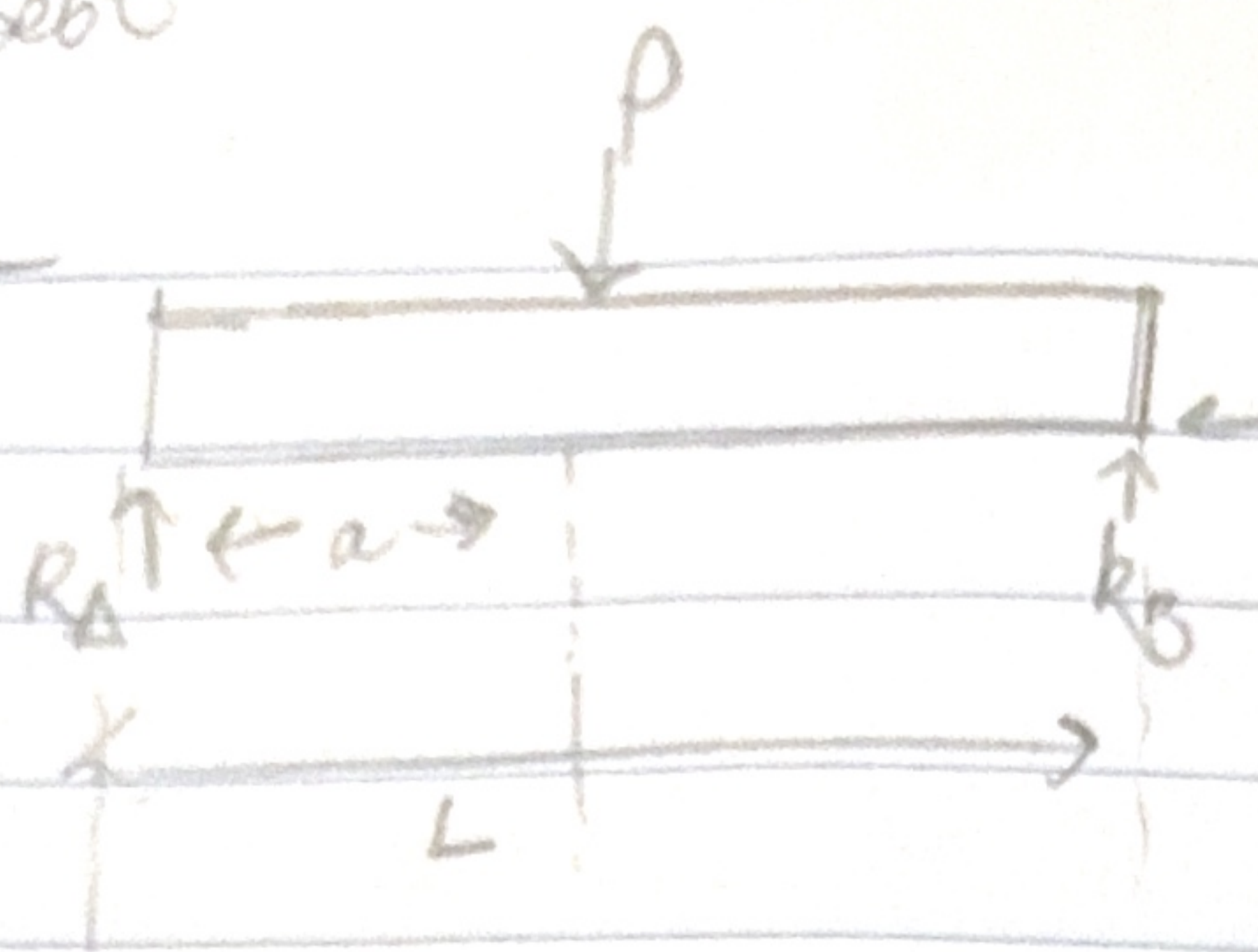
$$R_A + 5\text{kN} - 10\text{kN} = 0$$

$$R_A = 5\text{kN}$$

$$R_A = \frac{F}{2} = \frac{10}{2} = 5\text{kN}$$

$$R_B = \frac{F}{2} = \frac{10}{2} = 5\text{kN}$$

* simply supported
Structure 2



$$\sum M_B = 0$$

$$\sum M_B = R_A \times L - P(L-a) = 0$$

$$R_A \times L = P(L-a)$$

$$R_A = \frac{P(L-a)}{L}$$

$$\sum F_x = 0 \quad (\text{no external resultant})$$

$$\sum F_y = 0 \quad " \quad "$$

$$R_A + R_B - P = 0$$

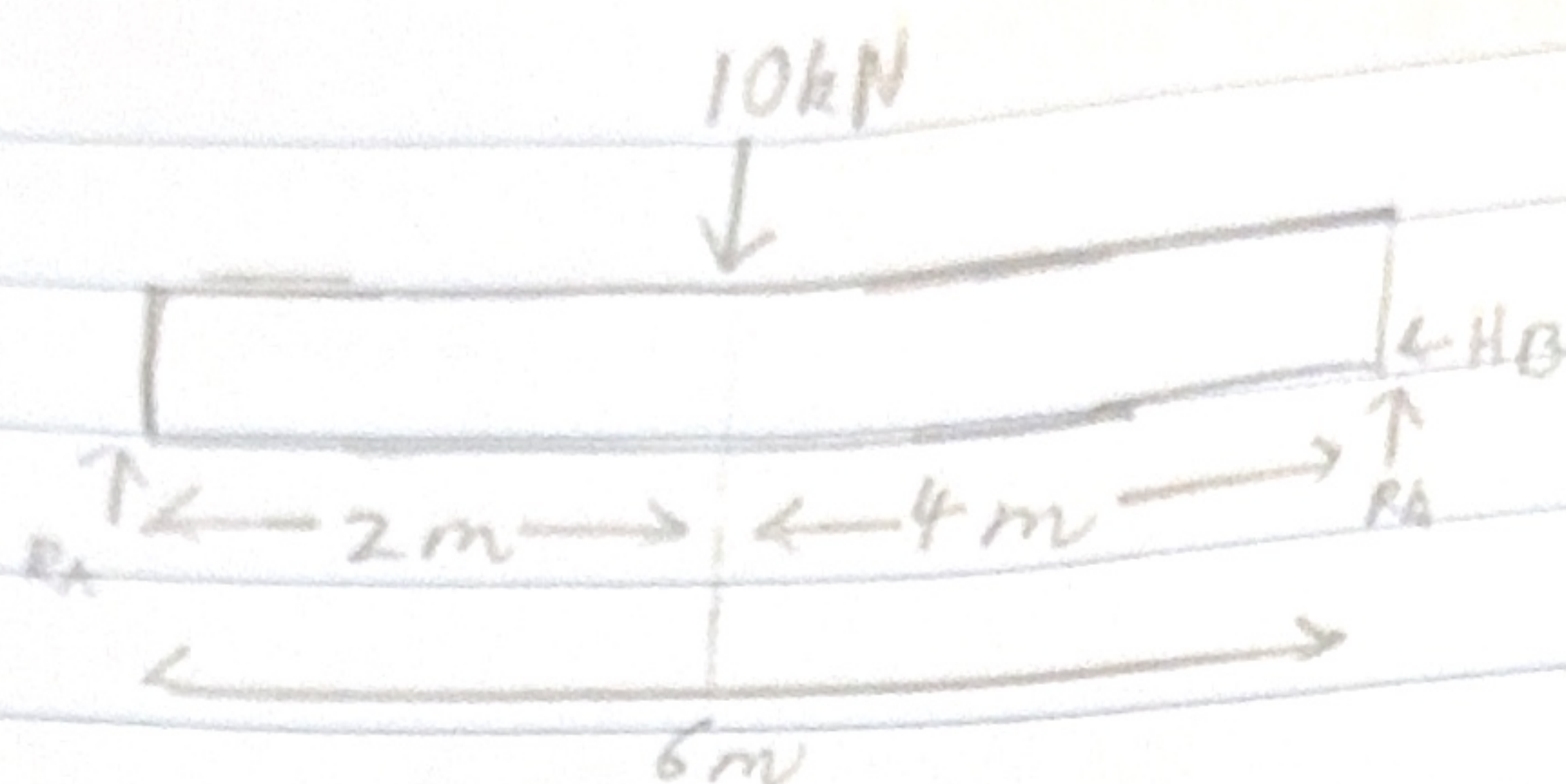
$$R_B = P - R_A$$

$$R_B = P - \frac{P(L-a)}{L}$$

$$R_B L = PL - PL + Pa$$

$$R_B L = Pa$$

$$R_B = \frac{Pa}{L}$$



$$\sum M_A = 0$$

$$\sum M_A \Rightarrow 10 \times 2 - R_B \times 6 = 0$$

$$\Rightarrow 20 - 6R_B = 0$$

$$\Rightarrow \frac{20}{6} = R_B = 3.33 \text{ kN}$$

$$\sum F_y = R_A + R_B - P = 0, \quad \sum F_x = 0$$

$$R_A + 3.33 - 10 = 0$$

$$R_A = 10 - 3.33 = 6.67$$

$$\sum M_B = R_A \times 6m - P \times 4m = 0$$

$$= 6.67 \times 6 - 10 \times 4$$

$$= 40.02 \text{ kNm} - 40 \text{ kNm}$$

$$= 0$$

$$R_A = \frac{P(L-a)}{L} \Rightarrow \frac{10(6-2)}{6} \Rightarrow 6.67 \text{ kN}$$

$$R_B = \frac{Pa}{L} \Rightarrow \frac{10 \times 2}{6} \Rightarrow 3.33 \text{ kN}$$